

Domain-Specific Languages of Mathematics

Course codes: DAT326 / DIT983

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Results	Announced within 15 workdays
Exam check	Through canvas
Aids	One textbook of your choice (Domain-Specific Languages of Mathematics, or Beta - Mathematics Handbook, or Rudin, or Adams and Essex, or ...). No printouts, no lecture notes, no notebooks, etc.
Grades	To pass you need a minimum of 5p on each question (1 to 4) and also reach these grade limits: 3: ≥ 48 p, 4: ≥ 65 p, 5: ≥ 83 p, max: 100p

Remember to write legibly. Good luck!

For reference: the learning outcomes. Some are tested by the hand-ins, some by the written exam.

- Knowledge and understanding
 - design and implement a DSL (Domain-Specific Language) for a new domain
 - organize areas of mathematics in DSL terms
 - explain main concepts of elementary real and complex analysis, algebra, and linear algebra
- Skills and abilities
 - develop adequate notation for mathematical concepts
 - perform calculational proofs
 - use power series for solving differential equations
 - use Laplace transforms for solving differential equations
- Judgement and approach
 - discuss and compare different software implementations of mathematical concepts

1. [25p] **Algebraic structure: a DSL for fields**

Consider the Wikipedia definition of a *field*:

Formally, a field is a set F together with two operations called addition and multiplication. [...] These operations are required to satisfy the following properties, referred to as field axioms. In the following definitions, a , b and c are arbitrary elements of the field F .

- **Associativity** of addition and multiplication: $a + (b + c) = (a + b) + c$ and $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
 - **Commutativity** of addition and multiplication: $a + b = b + a$ and $a \cdot b = b \cdot a$.
 - **Additive and multiplicative identity**: there exist two different elements 0 and 1 in F such that $a + 0 = a$ and $a \cdot 1 = a$.
 - **Additive inverses**: for every a in F , there exists an element in F , denoted $-a$, called additive inverse of a , such that $a + (-a) = 0$.
 - **Multiplicative inverses**: for every $a \neq 0$ in F , there exists an element in F , denoted by a^{-1} , $1/a$, or $\frac{1}{a}$, called the multiplicative inverse of a , such that $a \cdot a^{-1} = 1$.
 - **Distributivity** of multiplication over addition: $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$.
- (a) Define a type class *Field* that corresponds to the field structure (with operations *add*, *mul*, *zero*, *one*, *negate*, *recip*).
 - (b) Define a datatype $F\ v$ for the language of field expressions (with variables of type v) and define a *Field* instance for it. (These are expressions formed from applying the field operations to the appropriate number of arguments, e.g., all the left hand sides and right hand sides of the above equations.)
 - (c) Find and implement two other instances of the *Field* class.
 - (d) Give a type signature for, and define, a general evaluator for $F\ v$ expressions on the basis of an assignment function.
 - (e) Specialise the evaluator to the two *Field* instances defined in (1c). Take three field expressions of type $F\ String$, give the appropriate assignments and compute the results of evaluating, in each case, the three expressions.

Each question carries 5pts.

2. [25p] **Laplace**

Consider the following differential equation:

$$f''(t) + 4 \cdot f'(t) + 3 \cdot f(t) = -6 \cdot e^{-2t}, \quad f(0) = 0, \quad f'(0) = 12$$

- (a) [10p] Solve the equation assuming that f can be expressed by a power series fs , that is, use *integ* and the differential equation to express the relation between fs , fs' , fs'' . What are the first four coefficients of fs ? Explain how you compute them.
- (b) [15p] Solve the equation using the Laplace transform. You should need this formula (note that α can be a complex number) and the rules for linearity + derivative:

$$\mathcal{L}(\lambda t \cdot e^{\alpha * t}) s = 1/(s - \alpha)$$

Show that your solution does indeed satisfy the three requirements.

3. [25p] Typing maths: Partial derivative

Consider the following text from page 169 of Mac Lane [1968]:

[...] a function $z = f(x, y)$ for all points (x, y) in some open set U of the cartesian (x, y) -plane. [...] If one holds y fixed, the quantity z remains just a function of x ; its derivative, when it exists, is called the *partial derivative* with respect to x . Thus at a point (x, y) in U this derivative for $h \neq 0$ is

$$\frac{\partial z}{\partial x} = f'_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}$$

- (a) [5p] What are the types of the base sets and variables involved in the equation on the last line (z, U, x, y, h)?
- (b) [10p] What are the types of the functions and operators involved ($f, f'_x, \lim$, and the operator $\partial/\partial x$)?
- (c) [10p] Type and define a higher-order function that builds the argument to *lim* from f, x , and y .

You are welcome to introduce functions and names to explain your reasoning.

4. [25p] Equational reasoning and endomorphisms

Note: The learning outcome tested here is mainly “perform calculational proofs”, thus you are expected to use equational reasoning and motivate your steps.

An endohomomorphism on an additive group $(G, (+), 0)$ is a function from G to G that distributes over addition. In other words: $Endo(f) = H_2(f, (+), (+)) \wedge H_0(f, 0, 0)$ where $H_2(h, op_1, op_2) = \forall x. \forall y. h (op_1 x y) = op_2 (h x) (h y)$ and $H_0(h, e_1, e_2) = h e_1 = e_2$.

If the group is abelian (thus $(+)$ is commutative), endohomomorphisms form a ring with function composition for multiplication and pointwise addition:

$$\begin{aligned} mulE \ f_1 \ f_2 &= f_1 \circ f_2 && \text{-- Def. } mulE \\ addE \ f_1 \ f_2 &= lift2 \ (+) \ f_1 \ f_2 && \text{-- Def. } addE \\ f_1 \circ f_2 &= \lambda x \rightarrow f_1 (f_2 x) && \text{-- Def. } (\circ) \\ lift2 \ op \ f_1 \ f_2 &= \lambda x \rightarrow op (f_1 x) (f_2 x) && \text{-- Def. } lift2 \end{aligned}$$

- (a) [5p] Define the additive unit *zeroE* of this ring and prove that it is the right-additive unit.
- (b) [5p] Define the multiplicative unit *oneE* of this ring and prove that it is the right-multiplicative unit.
- (c) [15p] Prove $Endo(f) \Rightarrow (mulE \ f \ (addE \ g \ h) = addE \ (mulE \ f \ g) \ (mulE \ f \ h))$ for all f, g, h of appropriate type. (This proves the distributivity law for the ring instance.)